

Comparative Evaluation of Pre-Trained Models for Brain Tumor Identification based on MRI and CT Image

Atharva Daga¹, Viraj Laddha², Prathmesh Jain³, Tanmay Sharma⁴,

Department of Computer Engineering
Vishwakarma University Pune, India

Abstract- Brain tumor detection is important in neuroimaging, affecting patient outcomes and prognosis. To improve detection capabilities, this study uses MRI & CT Scan Image to classify brain tumor while employing deep learning techniques. We test how well pre-trained models like VGG-19, DenseNet-121, and ResNet-50 perform by using detailed information from MRI and CT scans to improve the accuracy of detecting brain tumors and help identify them more clearly and precisely, facilitating swift diagnosis and informed treatment planning. This research utilizes image fusion and prediction algorithms to address challenges such as limited data diversity and difficulties in differentiating tumor boundaries from surrounding tissues, thereby improving model performance. By evaluating the results, we identified the most accurate model for brain tumor diagnosis and provided insights into its use and impact on diagnosis. This research advances technology and improves patient outcomes through more accurate and timely diagnoses. Analysis shows Resnet-50 achieving the highest accuracy among all other models is effective for tumor detection.

Keywords – Brain tumour detection, Magnetic Resonance Imaging (MRIs), Computed Tomography (CT) scans, Image fusion, Pre – trained Models, and Image Analysis.

I. INTRODUCTION

Brain tumors are a big challenge for doctors to diagnose because they can lead to serious health problems, like damage to the nervous system and even life-threatening situations. Getting an accurate diagnosis quickly is very important for giving the best treatment and improving the chances of a good outcome for patients. Medical imaging plays a key role in diagnosing and seeing brain tumors, with CT scans and MRI being the main tools used. MRI is especially good at showing soft tissues, allowing doctors to see small differences in tissue structure that help in finding tumors. On the other hand, CT scans are better at showing hard tissues like bones and calcium deposits, giving extra information that complements what MRI provides. Using both MRI and CT together helps doctors get a clearer picture of brain tumors by combining the strengths of each imaging method. This helps make the detection of brain tumors more accurate and efficient, image fusion techniques are employed. Advanced methods such as wavelet transforms are used to combine MRI and CT images, preserving both high spatial resolution and essential diagnostic details. This fusion process mitigates the limitations of individual imaging techniques, thereby improving overall diagnostic performance.

Deep learning (DL) techniques are pivotal in enhancing brain tumor classification by automating the analysis of complicated

medical images, like MRI and CT scans, uses deep learning models. These models rely on neural networks, especially convolutional neural networks (CNNs), which excel in detecting subtle patterns in imaging data, resulting in more accurate and earlier tumor identification. This not only improves diagnostic precision but also promotes better patient outcomes through timely and personalized treatment options. This study looks at how well pre-trained models work when they analyze combined MRI and CT images to find brain tumors. The main goals are to: first, check how effectively these models can spot tumors in the merged images; second, compare their accuracy and effectiveness in detecting tumors; and third, determine the optimal approach for tumor detection by identifying the model that offers the highest precision and efficiency.

Our study looks at how image fusion affects the accuracy of detecting brain tumors and finds out what works best which pre-trained model offers the most precise evaluation of combined MRI and CT images. This study seeks to explore how combining these imaging techniques improves detection accuracy, while evaluating the performance of models in delivering accurate diagnostic results. This research is important because it has the potential to make diagnoses more accurate and help patients get better results by using advanced image fusion methods and evaluating how well pre-trained models work. The goal of this study is to help doctors make

more precise and quicker diagnoses, which in turn can lead to better choices in patient care and treatment plans.

The main purpose of this study is to detect brain tumors by combining MRI and CT scans. The research uses advanced image processing techniques and transfer learning to increase the accuracy and efficiency of tumor identification. The goal is to help doctors diagnose tumors earlier and take timely action, which can lead to better treatment results for patients.

II. LITERATURE REVIEW

The domain of brain tumor detection has experienced substantial progress recently, driven by the growing availability of imaging data and advances in computational capabilities. An array of methodologies has been investigated, ranging from conventional image processing techniques to advanced convolutional neural networks (CNNs) and transfer learning models. Researchers have employed various datasets and utilized numerous models, including decision trees, support vector machines (SVMs), neural networks, and sophisticated CNN architectures like VGG-16, ResNet, and DenseNet. The primary focus has often been on enhancing classification precision, refining segmentation accuracy, and improving the overall effectiveness of tumor detection systems. This review consolidates insights from several pivotal studies in the field, highlighting the datasets, models, and findings that have advanced brain tumor detection.

In the research of Dina Aboul Dahab, et al. applied image segmentation, Canny edge detection, and LVQ-based PNN techniques on a dataset of 64 MRI images for training and 18 for testing. The study introduced a modified PNN model, achieving a remarkable 100% classification accuracy with a spread value of 1 and a significant 79% reduction in processing time [10]. Similarly, by Vipin Y. Borole et al. offers a thorough overview of various preprocessing, feature extraction, segmentation, and post-processing techniques. It details several algorithms, including watershed segmentation, and reports tumor detection and segmentation accuracies exceeding 90% [12].

Janki Naik and Prof. Sagar Patel proposed a method that includes median filtering, texture feature extraction, and association rule mining. They used a dataset of 124 MRI images, with 73 for training and 51 for testing. The study applied Decision Tree and Naive Bayesian classifiers. The Decision Tree model had 100% precision, 93% sensitivity, 100% specificity, and 96% accuracy, while the Naive Bayesian classifier had 91% precision, 91% sensitivity, 83% specificity, and 88.2% accuracy [2]. In contrast, the study “Refining Neural Network Algorithms for Accurate Brain Tumor Classification in MRI Imagery” by Asma Alshuhail et al. focused on CNNs with Grad-CAM visualization and data augmentation. They

achieved an overall accuracy of 98%, with precision, recall, and F1- scores between 97% and 98%, and ROC-AUC values between 99% and 100% [16].

N. Varuna Shree and T. N. R. Kumar used a dataset of 650 samples from 25 images. They applied noise removal, GLCM feature extraction, DWT-based segmentation, and PNN classification. Their model achieved nearly 100% accuracy on the training set and 95% accuracy on the test set [17]. Similarly, Sandeep Kumar Mathivanan et al. used transfer learning with models like ResNet152, VGG19, DenseNet169, and MobileNetv3 on a dataset of 7,023 MRI images from Kaggle. They achieved a high accuracy of 99.75% using MobileNetv3 [18].

Eid Albalawi et al. combined federated learning and transfer learning with a modified VGG16 model. They tested it on 7,023 images from different sources and achieved 98% accuracy with high precision and recall across various tumor types [19]. Additionally, G. Hemanth et al. focused on preprocessing, data cleaning, and segmentation using CNN with the LinkNet architecture on MRI images from UCI and online resources. Their study reported 91% accuracy and 92.7% efficiency, which is better than existing techniques [6].

Falak Chhatre et al. used a dataset of 2,998 images from the Br35H dataset. Their study included preprocessing and CNN models (KSM, IAM), with IAM achieving 97.99% validation accuracy and 100% accuracy on the validation dataset [1]. Sunita M. Kulkarni and G. Sundari compared models like AlexNet, VGG16, ResNet18, ResNet50, and GoogLeNet on a dataset of 150 brain MRI images. AlexNet had the highest precision (0.937), recall (1), and F-measure (0.96774) [5].

Chetana Srinivas et al. explored transfer learning with VGG-16, ResNet-50, and Inception-v3 on a dataset of 233 MRI images (158 benign, 98 malignant). VGG-16 achieved 96% accuracy, 94% precision, 100% recall, and 98% F1-score [3]. Komal Sharma et al. used a dataset of 212 brain MRI images and applied grayscale conversion, noise removal, edge detection, watershed segmentation, and GLCM feature extraction. Multi-Layer Perceptron (MLP) achieved 98.6% accuracy, while Naive Bayes achieved 97.6% accuracy [13]. Javaria Amin and others used several models like SVM, KNN, Decision Tree, Random Forest, and Naive Bayes on local and BRATS datasets. Their study showed high accuracy across different datasets, with results based on pixels and high PSNR and SSIM scores [8]. Masoumeh Siar and Mohammad Teshnehlab used CNNs along with feature extraction and clustering on a dataset containing 1,892 MRI images. CNN combined with Softmax gave the best accuracy of 99.12%, and all classifiers had 100% sensitivity [9].

Dena Nadir George and others focused on extracting shape features using Sigma filtering and adaptive thresholding on 174 MRI samples.

They used the C4.5 decision tree and Multi-Layer Perceptron (MLP), achieving 95% precision with MLP and 91% precision with C4.5 [7]. On the other hand, Ankita Kadam and others did a comparison between ANN, CNN, and Transfer Learning models using a large dataset of 3,190 T1-weighted MRI images, which was expanded to 66,960 images. CNN achieved an accuracy of 94.15% [14].

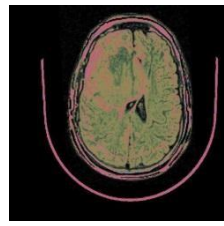
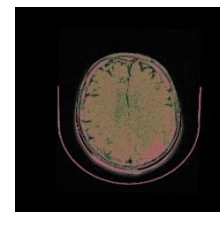
G.P. Hegde and others worked on image preprocessing, feature extraction, image fusion via wavelet transform, and segmentation using Otsu's algorithm. Even though they didn't specify the models used, their study found that fused images gave more accurate and dependable parameters than individual CT and MRI images. They analyzed parameters like mean, standard deviation, entropy, root mean square (RMS), variance, smoothness, kurtosis, skewness, inverse difference moment (IDM), contrast, correlation, energy, and homogeneity [15].

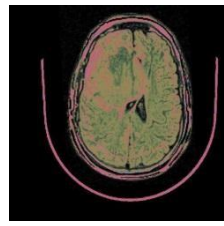
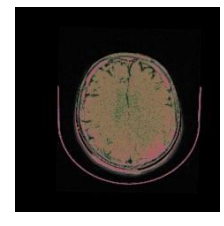
These studies collectively illustrate the dynamic and evolving field of brain tumor detection, integrating deep learning, machine learning, and traditional image processing techniques. The research highlights a wide range of models, from classical decision trees to cutting-edge CNNs, achieving high accuracies across various datasets. By utilizing diverse algorithms, transfer learning methods, and neural networks, the field continues to advance, enhancing detection capabilities and providing more precise classifications, which have significant implications for clinical practice.

III. DATA COLLECTION

The primary challenge in training deep learning models was acquiring a sufficiently large and diverse dataset to achieve precise results. To address this challenge, we utilize a comprehensive dataset from Kaggle, which includes fused MRI and CT scan segmented images with a resolution of 96 dpi and stored in .jpg format. This ensures uniformity and high quality throughout the dataset, which is crucial for developing models with robust generalization capabilities.

Table I: Dataset Description

Tumor	Number of Images (Fused MRI & CT Scan Images)	Sample Image
Yes	198	
No	155	

Yes	198	
No	155	

This varied dataset offers a strong basis for training and evaluating pre-trained models for medical imaging. However, because the Kaggle dataset was not large enough, we used image augmentation methods, as discussed earlier, to create more images. These techniques improved the training process by making the dataset more diverse. We allocated 80% of this augmented dataset for training, allowing the model to learn from a broad range of data, while the remaining 20% was set aside for rigorous validation and evaluation.

Our study seeks to advance detection capabilities by leveraging MRI to discern soft tissue and CT to assess tissue structure. Our proposed model undergoes training with a batch size of 64 over 50 epochs, focusing on binary classification into two categories: 'yes' (indicating the presence of a brain tumor) and 'no' (indicating its absence). Through a comprehensive evaluation, we identify the most precise model from the pre-trained models listed in Table I for brain tumor diagnosis and offer insights into its practical application and impact on diagnostic accuracy.

For the final assessment, we employ an independent dataset featuring real patient data, with a resolution of 96 dpi, provided in .jpg format, and comprising 11 patient cases. This separate dataset helps check how accurate and useful the model is in real-life situations. Using this organized method improved the model's ability to perform well and stay strong, which played a big role in creating dependable and efficient models for finding brain tumors using medical images.

IV. METHODOLOGY

Our study starts by using MRI and CT scan images. Given the limited number of images, we use data augmentation techniques using ImageDataGenerator to increase contrast and strength. This is a crucial process that aligns two different

images, such as MRI and CT scans, by transforming one image to match the coordinate system of the other. In this process, corresponding points from both images are manually selected and used to calculate an optimal transformation, which includes translation, rotation, and scaling to ensure precise alignment of images that may differ in size, orientation, or perspective.

This is typically done using the Procrustes function, which computes the transformation matrix. The matrix is then applied to the MRI image to align it with the CT image. Translation shifts the image to align corresponding points, rotation ensures that the images are oriented similarly, and scaling adjusts the image size to match. Image registration is necessary to ensure that the images are in the same spatial framework, enabling accurate comparison, analysis, and fusion of data from both scans. This alignment is important in medical imaging to help with accurate feature extraction, clear visualization, and improved decision-making, particularly when combining different modalities like CT for structural details and MRI for soft tissue information.

Many optimizations work, such as scaling, rotation, translation, shearing, zooming, horizontal flipping, and brightness modification.

Scaling:

$$I'(x, y) = I(sx, sy)$$

where s is the scaling factor.

Rotation:

$$I'(x, y) = I(sx, sy)$$

frequency band are combined using the VGG-19 network, which analyzes and integrates these features through its convolutional layers, ensuring the final fused image retains both coarse structural information and fine details. The output of the fusion process is a single image that where θ denotes the rotation angle. Translation:

$$I'(x, y) = I(x + tx, y + ty)$$

where t_x and t_y are the translation shifts. Brightness:

$$I'(x, y) = I(x, y) + b$$

where b is the brightness adjustment value.

V. FEATURE EXTRACTION

MRI and CT images were processed separately through a feature extraction phase to emphasize critical patterns. Each image undergoes wavelet decomposition, which segments it into four frequency bands: LL (Low-Low), LH (Low-High), LV (Low-Vertical), and LD (Low-Diagonal). These bands are mathematical constructs used for image fusion rather than

anatomical entities. The LL band contains essential low-frequency information representing the overall structure of the image, while the LH, LV, and LD bands capture horizontal, vertical, and diagonal edge details, respectively.

MRI and CT images are combined using a method that involves wavelets that utilizes the Resnet-50 network. This method begins with wavelet decomposition, breaking each image into the frequency bands: LL, LH, LV, and LD. The LL band contains the lowest frequency information and represents the primary structure and major features of the image. It provides a smooth, low-resolution approximation essential for preserving fundamental structural details. The LH band captures horizontal edge details and fine textures, including high-frequency components in the horizontal direction, crucial for identifying edges and texture patterns. The LV band highlights vertical edge details and fine textures, including high-frequency components in the vertical direction, aiding in detecting vertical structures and patterns. The LD band focuses on diagonal edge details and fine textures, including high-frequency components along diagonal directions, which are essential for recognizing diagonal structures and patterns in the image.

The fusion process integrates these frequency bands from MRI and CT images using the VGG-19 network. The network processes each frequency band separately, combining features from both modalities to produce a comprehensive representation. During the fusion, the LL band from both MRI and CT images is prioritized because it retains fundamental structural information crucial for accurate detection. The LH, LV, and LD bands are fused to capture detailed edge information, thereby enhancing the detection of fine features. Features extracted from each incorporates combining information from both MRI and CT scans provides a more detailed and thorough picture, which helps improve the accuracy of detecting brain tumors later on.

Following the fusion process, the final image is reconstructed from the fused frequency bands using inverse wavelet transformation. This step involves combining the frequency bands to form a unified image. The LL, LH, LV, and LD bands, fused separately, are combined to reconstruct the image. The inverse wavelet transformation merges these bands into a single image, integrating both low-frequency structural information and high-frequency details. The reconstruction process applies the inverse of wavelet decomposition, with the formula:

$$I'(x, y) = IDWTLL(x, y) + IDWTLH(x, y) + IDWTLV(x, y) + IDWTLD(x, y)$$

where "IDWT" denotes the Inverse Discrete Wavelet Transform for each band.

The inverse transformation combines contributions from the LL, LH, LV, and LD bands, resulting in a single, fused image. This final image integrates detailed features from both MRI and CT scans, providing a comprehensive view that leverages the strengths of both imaging modalities, combining detailed soft tissue contrasts from MRI with bone structure details from CT scans to enhance overall diagnostic quality.

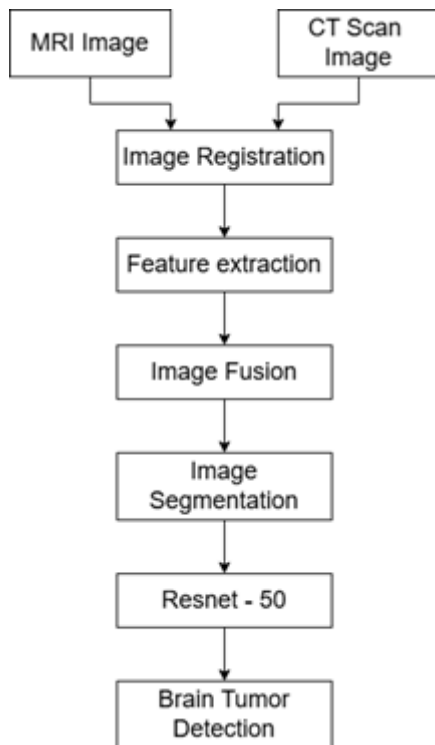


Fig.1. System Architecture

The combined images were used to train and test the model. Its performance was measured using metrics like precision, specificity, and sensitivity. The testing process compared different models to find out which one was best at detecting brain tumors.

VI. EXPERIMENTAL DESIGN

The main purpose of this study, shown in Fig. 1, is to identify brain tumors using a combination of MRI and CT scans. By using sophisticated image processing methods, the research aims to improve the accuracy and speed of tumor detection. The broader objective is to help in early diagnosis and timely treatment, which can lead to better results for patients.

A suite of Python libraries and frameworks is employed to support various tasks: OpenCV is used for image processing, NumPy handles numerical calculations, PyTorch facilitates deep learning, and Matplotlib is utilized for data visualization.

Additionally, Keras and TensorFlow are incorporated to enable effective image analysis and model training.

Pre-Trained Models:

- VGG-19:** Created by the Visual Geometry Group at the University of Oxford, VGG-19 is a deep convolutional neural network with 19 layers, of which 16 are convolutional and 3 are fully connected. This model is very good at capturing detailed information from MRI and CT scans, which makes it highly useful for spotting complex patterns that are important in detecting brain tumors.
- DenseNet-121:** DenseNet-121 features 121 layers and utilizes dense connections between layers to enhance feature propagation and reuse. This architecture effectively addresses the vanishing gradient issue and facilitates precise feature extraction, essential for distinguishing tumor from non-tumor tissues in MRI and CT scans.
- ResNet-50:** ResNet-50 uses a technique called residual learning, which includes shortcut connections, allowing it to stay accurate even though it has many layers. This model is good at recognizing complex patterns in medical images, helping to tell the difference between healthy and unhealthy brain tissues.

Combining MRI and CT images is done by changing them into the YCbCr color space, which splits the image into three parts: brightness (Y), and color information (Cb and Cr). This approach supports efficient compression and processing, leveraging human visual perception characteristics to optimize the storage and transmission of MRI and CT scan images in our model. A fusion strategy that involves softmax-weighted sums of these feature maps integrates data from the different images. The fused image is then reconstructed and saved. Additionally, wavelet transforms decompose the images into approximation and detail coefficients, which are merged at multiple levels to generate a final image. This technique effectively combines spatial and frequency domain features, enhancing the accuracy of brain tumor detection.

Watershed Algorithm

To refine segmentation accuracy, the Watershed Transform is applied. This method creates a topographical surface based on pixel intensities, with height values corresponding to intensity levels. The algorithm simulates flooding from local minima to split the image into separate areas based on differences in brightness. This method works well for clearly outlining the edges of tumors in detailed MRI and CT scans, which helps improve the accuracy of finding and separating tumors during analysis.

The dataset is split into 80% for training and 20% for testing, and cross-validation is used to make sure the model works well. The process of training and checking the model repeatedly helps prevent it from learning too much from the training data and ensures it can work well with new data. This detailed

method is meant to improve accuracy in detecting brain tumors and help doctors make decisions quickly when needed.

VII. RESULTS

The study evaluated VGG-19, ResNet-50, and DenseNet-121 for brain tumor detection using fused MRI and CT images. Resnet-50 achieved the highest accuracy at 90% listed in Table II, followed by Densenet-121 and VGG-19. Image fusion improved detection by combining MRI and CT strengths, and data augmentation enhanced model performance. According to Fig.3. & Fig.4., Resnet- 50's superior accuracy and precision highlight its effectiveness in detecting brain tumors, supporting more accurate and timely diagnoses.

Table II: Results

MODEL	PRECISION	SENSITIVITY	SPECIFICITY	ACCURACY
VGG-19	87	84	84	82
DENSENET-121	82.4	89.8	86.2	87.5
RESNET-50	84	100	100	90



Fig.3. Accuracy over Epochs for Resnet-50 model

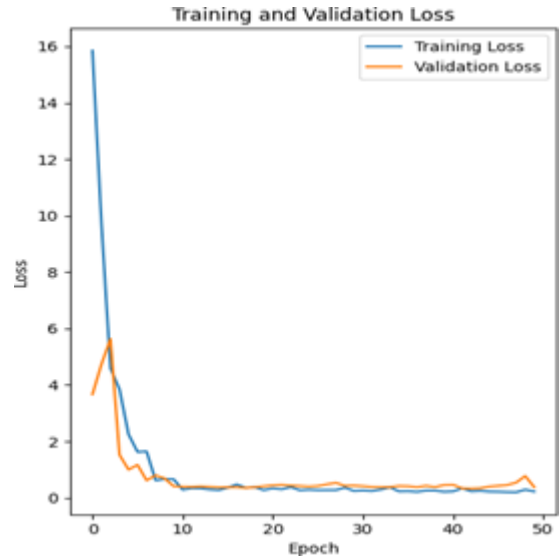


Fig.4. Loss over Epochs for Resnet-50 model

VIII. CONCLUSION

This research helps detect brain tumors more accurately by combining MRI and CT scans using advanced image fusion and deep learning methods. By merging the detailed soft tissue information from MRI with the structural details from CT, the study improves the precision of identifying tumors. Evaluating pre-trained models (VGG- 19, DenseNet-121, and ResNet-50) revealed their strengths in different aspects of detection, with data augmentation and the Watershed Algorithm further refining accuracy. Overall, these methods advance diagnostic precision and support timely, accurate tumor identification, offering a promising direction for improving patient outcomes. Future research should expand datasets and explore additional models to enhance clinical applicability.

REFERENCES

1. Falak Chhatre, Sudhanva Deshpande, Sidhant Malik, Grace Yan & Suresh Subramaniam, "Brain Tumor Detection Using Convolutional Neural Network." Journal of Student Research Volume 12 Issue 2 (2023)
2. Janki Naik, & Prof. Sagar Patel, "Tumor Detection and Classification using Decision Tree in Brain MRI." Tumor Detection and Classification using Decision Tree in Brain MRI | ISSN: 2321-9939.
3. Chetana Srinivas, Nandini Prasad KS, Mohammed Zakariah, Yousef Ajmi Alothaibi, Kamran Shaukat, B Partibane, & Halifa Awal, "Deep transfer learning approaches in performance analysis of brain tumor classification using MRI images." Journal of Healthcare Engineering Volume 2022, Article ID 3264367, 17 pages.
4. Anil, Abhishek, Aditya Raj, H. Aravind Sarma, and NCRD PL. "Brain Tumor detection from brain MRI using Deep

- Learning." International Journal of Innovative Research in Applied Sciences and Engineering (IJIRASE) 3, no. 2 (2019): 458-465.
5. Sunita M Kulkarni, G Sundari, "Comparative analysis of performance of Deep CNN based framework for brain MRI classification using transfer learning." Journal of Engineering Science and Technology Vol. 16, No. 4 (2021) 2901 – 2917.
6. G Hemanth, M Janardhan, L Sujihelen, "Design and implementing brain tumor detection using machine learning approach." Proceedings of the Third International Conference on Trends in Electronics and Informatics (ICOEI 2019)
7. Dena Nadir George, Hashem B Jehlol, & Anwer Subhi Abdulhussein Oleiwi "Brain tumor detection using shape features and machine learning algorithms." International Journal of Scientific & Engineering Research, Volume 6, Issue 12, December-2015 ISSN 2229-5518
8. Javaria Amin, Muhammad Sharif, Mudassar Raza, Tanzila Saba, & Muhammad Almas Anjum, "Brain tumor detection using statistical and machine learning method." Computer methods and programs in biomedicine S0169-2607(18)31378-6, COMM4924
9. Masoumeh Siar, Mohammad Teshnehlab, "Brain tumor detection using deep neural network and machine learning algorithm." 9th International Conference on Computer and Knowledge Engineering (ICCKE 2019), October 24-25 2019
10. Dina Aboul Dahab, Samy S. A. Ghoniemy, & Gamal M. Selim, "Automated Brain Tumor Detection and Identification Using Image Processing and Probabilistic Neural Network Techniques." International Journal of Image Processing and Visual Communication ISSN 2319-1724: Volume 1, Issue 2, October 2012.
11. Pratibha Sharma, Manoj Diwakar, & Sangam Choudhary, "Application of Edge Detection for Brain Tumor Detection." International Journal of Computer Applications (0975 – 8887) Volume 58–No.16, November 2012.
12. Vipin Y. Borole, Sunil S. Nimbhore, & Dr. Seema S. Kawthekar, "Image Processing Techniques for Brain Tumor Detection: A Review." International Journal of Emerging Trends & Technology in Computer Science (IJETTCS), Volume 4, Issue 5(2), September - October 2015.
13. Komal Sharma, Akwinder Kaur, Shruti Gujral, "Brain tumor detection based on machine learning algorithms." International Journal of Computer Applications (0975 – 8887) Volume 103 – No.1, October 2014.
14. Ankita Kadam, Sartaj Bhuvaji, Sujit Deshpande, "Brain Tumor Classification using Deep Learning Algorithms." International Journal for Research in Applied Science & Engineering Technology (IJRASET) Volume 9 Issue XII Dec 2021.
15. G.P.Hegde, Chethana R Shetty, Chinmaye K, Roshani N G, "Segmentation Based Detection of Brain Tumor using CT, MRI and Fused Images." Special Issue - 2018 International Journal of Engineering Research & Technology (IJERT) ISSN: 2278-0181 ICRTT - 2018 Conference Proceedings
16. Asma Alshuhail, Arastu Thakur, R Chandramma, T R Mahesh, Ahlam Almusharraf, V Vinoth Kumar and Surbhi Bhatia Khan "Refining neural network algorithms for accurate brain tumor classification in MRI imagery." BMC Medical Imaging (2024)
17. N. Varuna Shree, T. N. R. Kumar, "Identification and classification of brain tumor MRI images with feature extraction using DWT and probabilistic neural network." Brain Informatics (2018)
18. Sandeep Kumar Mathivanan, Sridevi Sonaimuthu, Sankar, Murugesan, Hariharan Rajadurai, Basu Dev Shivahare & Mohd Asif Shah, "Employing deep learning and transfer learning for accurate brain tumor detection." Sci Rep 14, 7232 (2024).
19. Eid Albalawi, Mahesh T.R., Arastu Thakur, V. Vinoth Kumar, Muskan Gupta, Surbhi Bhatia Khan and Ahlam Almusharraf, "Integrated approach of federated learning with transfer learning for classification and diagnosis of brain tumor." BMC Medical Imaging, 2024